

Numerical Integration

MATH 146 – Calculus II

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Recall: Fundamental Theorem of Calculus

Theorem: Fundamental Theorem of Calculus

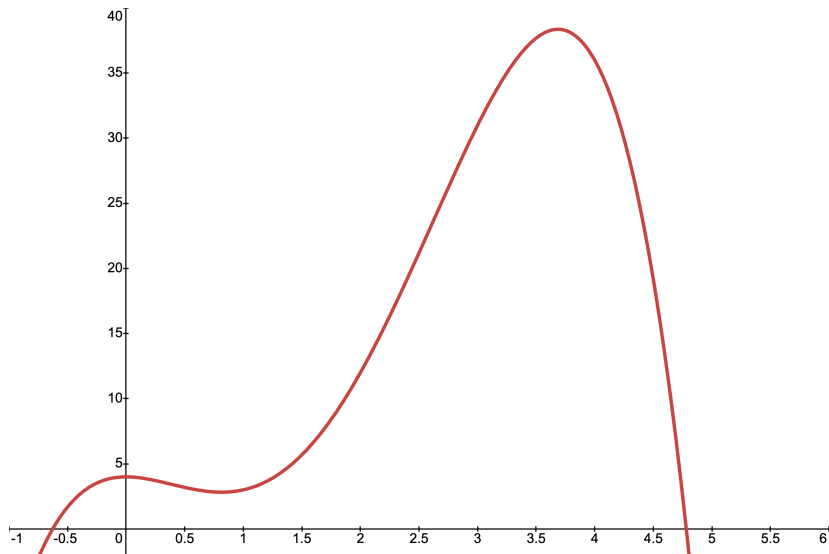
If $f(x)$ is continuous on $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

If this doesn't work out, we can approximate $\int_a^b f(x) dx$ using endpoint or midpoint approximations.

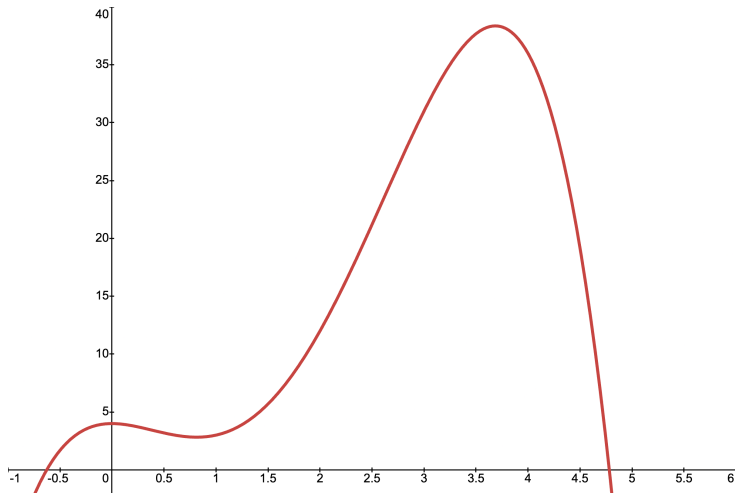
Is this the best option? What are the flaws to this approach?

Recall: Fundamental Theorem of Calculus



The Trapezoid Rule

What if instead of rectangles, we use trapezoids?



The Trapezoid Rule

Trapezoid Rule

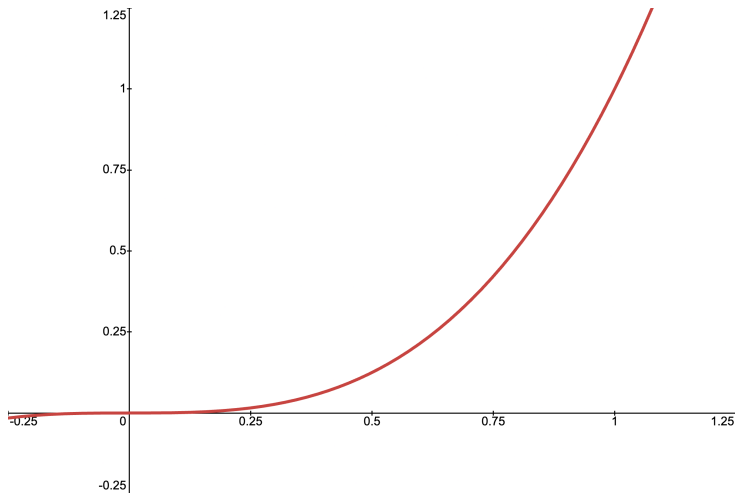
If $f(x)$ is a function on the interval $[a, b]$ and if n is a positive number, then the trapezoid approximation T_n for $\int_a^b f(x) dx$ with n subintervals is

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \left(f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n) \right),$$

where $x_j = a + j \left(\frac{b-a}{n} \right)$.

Example: The Trapezoid Rule

Example: Approximate $\int_0^1 x^3 dx$ with $n = 4$.



Approximating with Parabolas

Suppose (x_0, y_0) , (x_1, y_1) and $\left(\frac{x_0 + x_1}{2}, y_2\right)$ are three points in the plane. What is the area under the parabola $p(x) = ax^2 + bx + c$ passing through these points?



Simpson's Rule

Simpsons Rule

If $f(x)$ is a function on the interval $[a, b]$ and if $2n$ is an *even* positive number, then the Simpson approximation S_{2n} for $\int_a^b f(x) dx$ with $2n$ subintervals is

$$\int_a^b f(x) dx \approx \frac{b-a}{6n} \left(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) \right. \\ \left. + \cdots + 2f(x_{2n-2}) + 4f(x_{2n-1}) + f(x_n) \right),$$

where $x_j = a + j \left(\frac{b-a}{2n} \right)$.

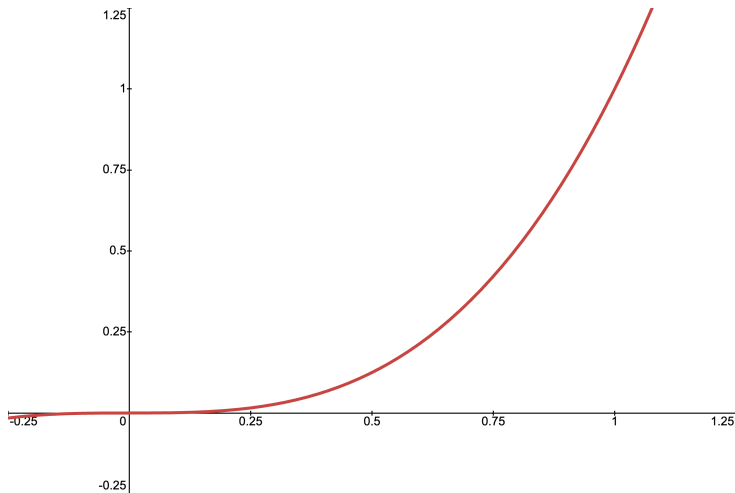
Simpson's Rule

Note: If $k = 2n$, then

$$\int_a^b f(x) dx \approx \frac{b-a}{6n} \left(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) \right. \\ \left. + \cdots + 2f(x_{k-2}) + 4f(x_{k-1}) + f(x_k) \right).$$

Example: The Simpson's Rule

Example: Approximate $\int_0^1 x^3 dx$ with $n = 4$.



Errors in Numerical Integration

Errors in Simpson's Method

If S_n is the estimate for $\int_a^b f(x) dx$ from Simpson's method, then the error E_s in Simpson's method is given by

$$E_s = \left| S_n - \int_a^b f(x) dx \right|.$$

Errors in Numerical Integration

Theorem (Errors in Numerical Integration)

Assume f'' is continuous on the interval $[a, b]$ and k is a real number such that $|f''(x)| \leq k$ for all x in $[a, b]$. The absolute errors in approximating the integral $\int_a^b f(x) dx$ by the Midpoint Rule and Trapezoid Rule with n subintervals satisfy the inequalities

$$E_M \leq \frac{k(b-a)}{24}(\Delta x)^2 \quad \text{and} \quad E_T \leq \frac{k(b-a)}{12}(\Delta x)^2,$$

respectively, where $\Delta x = \frac{b-a}{n}$.

Errors in Numerical Integration

Theorem (Errors in Numerical Integration)

Assume $f^{(4)}$ is continuous on the interval $[a, b]$ and K is a real number such that $|f^{(4)}(x)| \leq K$ for all x in $[a, b]$. The absolute error in approximating the integral $\int_a^b f(x) dx$ by Simpson's Rule with n subintervals satisfy the inequality

$$E_S \leq \frac{k(b-a)}{180}(\Delta x)^4,$$

where $\Delta x = \frac{b-a}{n}$.

Example: The Trapezoid Rule

Example: Estimate the error from approximating $\int_0^1 x^3 dx$ using T_4 .

Example: How large must n be so that $E_T \leq 0.00001$?